

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Computer Networks

COURSE CODE: CSA0702

COURSE OUTCOME COVERED

CO5: *Measure network performance using key metrics with simulation tools*

Assessment Tool Weightage: 20%

QUESTIONS: 40

Total Marks:40

Annexure A

Mock Interview

Code of Conduct:

I K.Yamini (192572181) (Name / Reg No) *certify* that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K. Y.

QUESTIONS:40

Total Marks:40

1. Analyze how network congestion affects packet delay and packet loss. (BL4)(1)
2. Compare the impact of star and mesh topologies on network reliability and fault tolerance. (BL4)(1)
3. Why does increasing the number of routers between source and destination increase end-to-end delay? Explain logically. (BL4)(1)
4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic. (BL4)(1)
5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication. (BL5)(1)
6. Analyze how network segmentation using VLANs improves performance and security. (BL4)(1)
7. Compare the effects of TCP and UDP on network performance for different applications. (BL4)(1)

8. Why is load balancing important in large enterprise networks? Analyze its impact on performance and availability. (BL5)(1)
9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization. (BL4)(1)
10. Evaluate the trade-off between network security measures (such as firewalls and encryption) and overall network performance. (BL5)(1)
11. A network has high delay—how will you identify and fix the issue? (BL4)(1)
12. If throughput is low despite high bandwidth, what could be the reasons? (BL4)(1)
13. How would you reduce packet loss in a congested network? (BL4)(1)
14. A VoIP call is breaking frequently—how would you troubleshoot? (BL4)(1)
15. Given a network with high jitter, suggest improvements (BL4)(1)
16. How will you simulate a congested network in Packet Tracer? (BL5)(1)
17. If packets are delayed in a router, what steps will you take? (BL4)(1)
18. How would you optimize a network for online gaming? (BL4)(1)
19. Design a small network and explain how you will measure its performance (BL5)(1)
20. If RTT is very high, how will you diagnose the issue? (BL4)(1)
21. Explain network performance to a non-technical person (BL5)(1)
22. Describe throughput using a real-life example (BL5)(1)
23. Present how packet loss affects video streaming (BL5)(1)
24. Explain delay types clearly in under 1 minute (BL4)(1)
25. Describe Cisco Packet Tracer in a professional way (BL5)(1)
26. Explain QoS in simple and technical terms (BL5)(1)
27. How would you present your simulation results to a client? (BL4)(1)
28. Explain jitter using a real-world analogy (BL5)(1)
29. Describe how you tested network performance in a lab (BL4)(1)
30. Explain the importance of performance metrics in one structured answer (BL5)(1)
31. Introduce yourself and explain your knowledge of network performance (BL5)(1)
32. Why are you interested in networking and simulation tools? (BL5)(1)
33. What makes you confident in analyzing network performance? (BL5)(1)
34. Explain a concept you are very strong in (related to networking) (BL5)(1)
35. How would you handle a question you don't know? (BL5)(1)
36. Describe a situation where you solved a technical problem (BL4)(1)
37. Why should we select you for a networking role? (BL5)(1)
38. What challenges did you face while learning Packet Tracer? (BL5)(1)
39. How do you improve your technical skills regularly? (BL5)(1)
40. Demonstrate confidence while explaining a complex concept simply (BL5)(1)

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding ; answers all questions accurately	Shows good understanding with few errors	Partial understanding ; several incorrect responses	Lacks understanding ; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

1. Analyze how network congestion affects packet delay and packet loss.

Network congestion occurs when too much traffic enters the network. Routers become overloaded, causing packets to wait in queues, which increases **delay**. When buffers become full, packets are dropped, causing **packet loss**. Therefore, congestion reduces overall network performance.

2. Compare the impact of star and mesh topologies on network reliability and fault tolerance.

A **star topology** is easy to manage, but failure of the central switch can affect the entire network. A **mesh topology** has multiple paths between devices, so if one link fails, data can use another path. Therefore, mesh provides higher reliability and fault tolerance, while star is simpler and cheaper.

3. Why does increasing the number of routers between source and destination increase end-to-end delay?

Each router introduces **processing delay, queuing delay, and transmission delay**. When the number of routers increases, these delays accumulate. Therefore, more hops generally result in higher end-to-end delay.

4. Analyze how QoS improves the performance of voice and video traffic.

Quality of Service (QoS) gives priority to delay-sensitive traffic such as voice and video. It manages bandwidth and reduces delay, jitter, and packet loss. This provides smoother video and clearer voice communication.

5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication.

Fiber optic cables provide **higher bandwidth, lower signal loss, and longer transmission distances** than copper. They are also less affected by electromagnetic interference. Although fiber has a higher installation cost, it is better for high-speed and long-distance communication.

6. Analyze how network segmentation using VLANs improves performance and security.

VLANs divide a physical network into separate logical networks. This reduces unnecessary broadcast traffic and improves network performance. VLANs also isolate users and departments, improving security by limiting direct communication between different groups.

7. Compare the effects of TCP and UDP on network performance for different applications.

TCP provides reliable and ordered delivery but has more overhead, so it can be slower. **UDP** has lower overhead and is faster but does not guarantee delivery. TCP is suitable for web and file transfer, while UDP is useful for streaming, gaming, and VoIP.

8. Why is load balancing important in large enterprise networks?

Load balancing distributes network traffic across multiple servers or links. It prevents one resource from becoming overloaded and improves response time. It also increases availability because traffic can be redirected if one server or link fails.

9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization.

Routing updates help routers maintain accurate network paths. However, very frequent updates consume bandwidth and router processing resources. Therefore, routing updates should be optimized to maintain accurate routing without creating unnecessary overhead.

10. Evaluate the trade-off between network security measures and overall network performance.

Firewalls, encryption, and security inspection improve protection but require additional processing. This can slightly increase delay and reduce throughput. However, properly configured security provides a good balance between protection and performance.

11. A network has high delay—how will you identify and fix the issue?

I would use **ping and traceroute** to identify where the delay occurs. Then I would check bandwidth usage, router queues, congestion, and faulty links. I would reduce congestion, optimize routing, or upgrade the affected link.

12. If throughput is low despite high bandwidth, what could be the reasons?

Possible reasons include **packet loss, congestion, high latency, interference, protocol overhead, faulty devices, or poor configuration**. High bandwidth only represents the maximum capacity; actual throughput depends on network conditions.

13. How would you reduce packet loss in a congested network?

I would identify the congested link using monitoring tools and reduce unnecessary traffic. I could apply QoS, increase bandwidth, optimize routing, and use load balancing. These methods help reduce congestion and packet drops.

14. A VoIP call is breaking frequently—how would you troubleshoot?

I would check **packet loss, delay, jitter, bandwidth, and connectivity**. I would use ping and network monitoring tools to identify the problem. Applying QoS and reducing congestion can improve VoIP quality.

15. Given a network with high jitter, suggest improvements.

I would identify congestion and inconsistent traffic flow. Then I would apply **QoS, traffic prioritization, buffering, and load balancing**. Increasing available bandwidth can also help reduce jitter.

16. How will you simulate a congested network in Packet Tracer?

I would create several network devices and generate high traffic between multiple nodes. I would configure limited-bandwidth links and observe packet transmission. I would then measure delay, packet loss, and throughput before and after applying improvements.

17. If packets are delayed in a router, what steps will you take?

First, I would check router CPU and memory utilization. Then I would check interface statistics, queue length, bandwidth, and packet loss. Finally, I would optimize routing, reduce unnecessary traffic, or upgrade the router/link if required.

18. How would you optimize a network for online gaming?

I would minimize **latency, jitter, and packet loss**. I would use QoS to prioritize gaming traffic, select efficient routes, avoid network congestion, and use a stable high-speed connection.

19. Design a small network and explain how you will measure its performance.

I would connect PCs to a switch, connect the switch to a router, and connect the router to another network. I would measure **delay using ping, path using traceroute, packet loss using repeated ping, and throughput using traffic-generation or monitoring tools**.

20. If RTT is very high, how will you diagnose the issue?

I would use **ping** to measure RTT and **traceroute** to identify the problematic hop. Then I would check congestion, routing paths, bandwidth, and faulty links. Correcting the route or reducing congestion can lower RTT.

21. Explain network performance to a non-technical person.

Network performance means **how well a network delivers data**. For example, a good network loads websites quickly, supports clear video calls, and does not frequently disconnect. We measure it using speed, delay, packet loss, and reliability.

22. Describe throughput using a real-life example.

Throughput is the **actual amount of data successfully transferred in a given time**. For example, if a road can handle 100 cars per minute but only 70 cars actually pass, the actual flow is similar to network throughput.

23. Present how packet loss affects video streaming.

packet loss means some video data does not reach the receiver. This can cause **freezing, buffering, reduced quality, or missing video frames**. Higher packet loss generally results in a poor streaming experience.

24. Explain delay types clearly in under 1 minute.

There are four main types of network delay: **processing delay, queuing delay, transmission delay, and propagation delay**. Processing is the time taken to examine a packet. Queuing is the waiting time in a buffer. Transmission is the time to place the packet onto the link. Propagation is the time for the signal to travel through the medium.

25. Describe Cisco Packet Tracer in a professional way.

Cisco Packet Tracer is a **network simulation and learning tool** used to design, configure, and troubleshoot networks virtually. It allows users to work with routers, switches, PCs, and network protocols without requiring physical hardware.

26. Explain QoS in simple and technical terms.

In simple terms, QoS means **giving important traffic higher priority**. Technically, QoS manages network traffic using techniques such as classification, prioritization, queuing, and bandwidth allocation to control delay, jitter, and packet loss.

27. How would you present your simulation results to a client?

I would present the results using **tables and simple graphs**. I would show important metrics such as throughput, delay, packet loss, and jitter. Finally, I would compare the results with the required performance and provide recommendations.

28. Explain jitter using a real-world analogy.

Jitter is the **variation in packet arrival time**. For example, if buses are expected every 10 minutes but arrive after 5, 15, and 8 minutes, their arrival is inconsistent. Similarly, variable packet arrival times create network jitter.

29. Describe how you tested network performance in a lab.

I created a network topology using network devices and configured communication between nodes. I generated traffic and measured **delay, packet loss, throughput, and jitter**. I then analyzed the results and identified possible performance improvements.

30. Explain the importance of performance metrics in one structured answer.

Network performance metrics help determine how efficiently a network operates.

Throughput measures successful data transfer, **delay** measures delivery time, **packet loss** measures lost packets, and **jitter** measures variation in delay. These metrics help identify problems and improve network quality.

31. Introduce yourself and explain your knowledge of network performance.

Good morning. I am **K. Yamini**, a Computer Science student. I have learned networking concepts such as TCP/IP, routing, switching, network topologies, QoS, and performance metrics. I also have basic experience with network simulation using Cisco Packet Tracer.

32. Why are you interested in networking and simulation tools?

I am interested in networking because it explains how devices communicate and share information. Simulation tools such as Cisco Packet Tracer allow me to **design, test, troubleshoot, and analyze networks** without requiring physical equipment.

33. What makes you confident in analyzing network performance?

My understanding of metrics such as **delay, throughput, packet loss, jitter, and RTT** gives me confidence. I can use tools such as ping, traceroute, and Packet Tracer to identify performance problems and suggest suitable solutions.

34. Explain a concept you are very strong in related to networking.

I am strong in **network performance analysis**. I understand how delay, throughput, packet loss, and jitter affect communication. By measuring these metrics, I can identify congestion and other network problems.

35. How would you handle a question you don't know?

I would remain calm and honestly say that I do not know the exact answer. I would explain what I know related to the topic and then try to logically analyze the question. I would also learn the correct answer afterward.

36. Describe a situation where you solved a technical problem.

During network simulation, if devices could not communicate, I would systematically check the **IP addresses, subnet masks, default gateways, cables, interfaces, and routing configuration**. By checking each layer step by step, I can identify and correct the configuration problem.

37. Why should we select you for a networking role?

I have a good foundation in networking concepts and performance analysis. I am willing to learn new technologies, troubleshoot problems systematically, and work responsibly. I can also use simulation tools to understand and test network configurations.

38. What challenges did you face while learning Packet Tracer?

Initially, configuring IP addresses, routing, and device interfaces was challenging. I overcame this by practicing small network topologies and troubleshooting configuration errors step by step. Regular practice improved my confidence.

39. How do you improve your technical skills regularly?

I improve my skills through **regular practice, simulations, lab exercises, technical documentation, and solving networking problems**. I also review mistakes and learn from them to improve my understanding.

40. Demonstrate confidence while explaining a complex concept simply.

A complex concept can be explained by breaking it into simple parts. For example, **QoS is like a priority system on a road**: emergency vehicles are given priority so they reach their destination faster. Similarly, QoS prioritizes important network traffic such as voice and video to reduce delay and improve quality.

